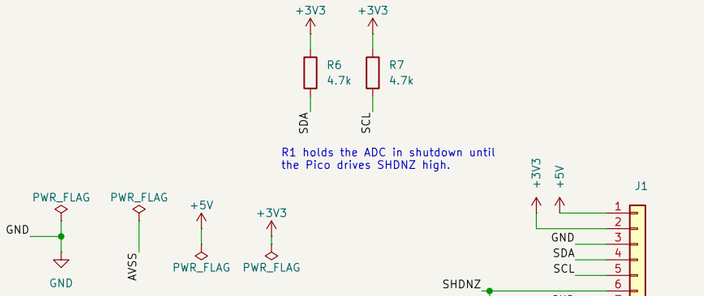
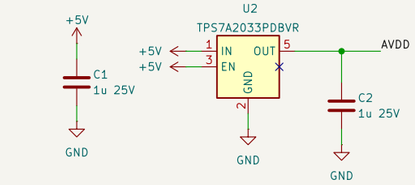
TLV320ADC6140 breakout for Stage 2 of TEST_PLAN_MCU_CAPTURE.md. It connects to a Pico 2 by jumper wires.

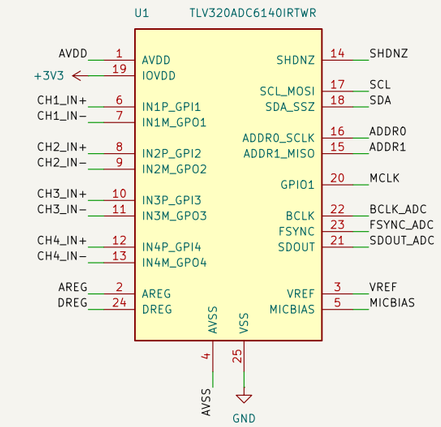
POWER AND CONTROL



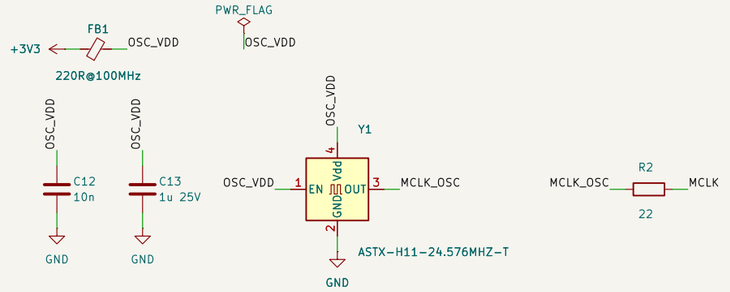
AVDD REGULATOR



ADC

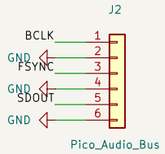


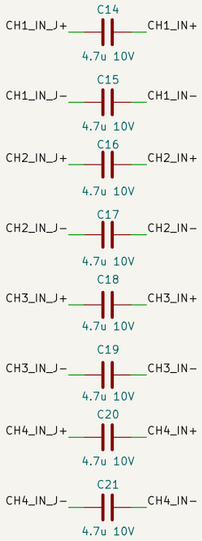
AUDIO REFERENCE 24.576 MHz



Standby pin 1 is tied to VDD, so the TCXO always runs.
MCLK enters the ADC on GPIO1.

AUDIO BUS (ADC is bus master)



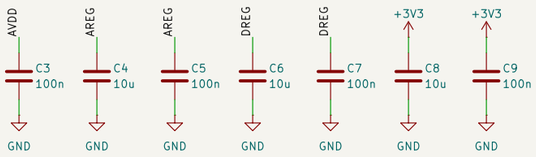


Analog Inputs

File: analog_inputs.kicad_sch

C14–C21 are 4.7 uF X7R, as on Ensor.
Set the ADC inputs to 10 kohm for a 3.4 Hz corner.
With 4.7 uF, firmware must lengthen the input quick-charge time.

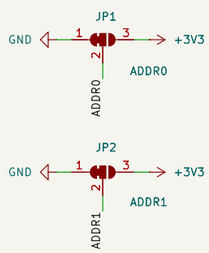
ADC DECOUPLING. Place each 100 nF cap at its pin.



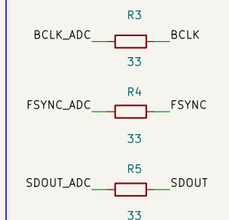
TEST POINTS



I2C ADDRESS

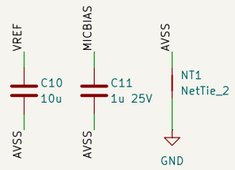


Default: both bridged to GND, address 0x4C.
Cut 1–2 and bridge 2–3 to change it.



R3–R5 sit next to U1.
Swap for 0 ohm if not needed.

C10 & C11:
Route ground terminal straight to AVSS, no via.



C11:
There are no onboard mics, so MICBIAS is not used.
Populate C11 for decoupling in case of activation in firmware.

Stage 2 test board. Parts follow docs/bom.md, ADC breakout section.

Portable Float Recorder

Sheet: /
File: adc-breakout.kicad_sch

Title: TLV320ADC6140 ADC Breakout

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/2

Sheet: /Analog Inputs/
File: analog_inputs.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/2